

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Revision Test 2026

MATHEMATICS — SSC-I (9th Grade) — ANSWER KEY (Version 18)

Syllabus: Unit 1 – Real Numbers, Unit 2 – Logarithms, Unit 4 – Factorization and Algebraic Manipulation, Unit 5 – Linear Equations and Inequalities, Unit 7 – Coordinate Geometry

Total Marks: 50

Time Allowed: 2 Hours 15 Minutes

Version 18

SECTION A — Answers to Multiple Choice Questions ($10 \times 1 = 10$ Marks)

Q1. $\sqrt{98} + \sqrt{8}$ is equal to:

Correct answer: **B) $9\sqrt{2}$** 1 mark

Simplify each radical first, then add like terms: $\sqrt{98} = \sqrt{(49 \times 2)} = 7\sqrt{2}$ and $\sqrt{8} = \sqrt{(4 \times 2)} = 2\sqrt{2}$, so the sum is $9\sqrt{2}$.

Adding the numbers under the radical signs ($98 + 8 = 106$) is **not** allowed: $\sqrt{a} + \sqrt{b} \neq \sqrt{a + b}$.

Source in her textbook: Unit 1, radicals (product rule $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$) — SLO: explain the concept of radicals and radicands; apply the laws of exponents. *Re-test of exam-17 Q2 ($\sqrt{48} + \sqrt{27}$ answered as 75).*

Q2. $\log 6 + \log 5$ is equal to:

Correct answer: **A) $\log 30$** 1 mark

By the **product law**, $\log m + \log n = \log(mn)$. So $\log 6 + \log 5 = \log(6 \times 5) = \log 30$. Adding logs multiplies the numbers; it does not add them ($\log 11$ is wrong).

Source in her textbook: Unit 2, laws of logarithm (product law) — SLO: prove the four basic laws of logarithm and apply them.

Q3. $\log_3 (1/27)$ is equal to:

Correct answer: **C) -3** 1 mark

$\log_3 (1/27) = x$ means $3^x = 1/27 = 3^{-3}$, so $x = -3$. The value is negative because $1/27$ is less than 1.

Source in her textbook: Unit 2, logarithm of a number to the base a — SLO: define logarithm of a number to the base a .

Q4. The factors of $27m^3 + 64n^3$ are:

Correct answer: **D) $(3m + 4n)(9m^2 - 12mn + 16n^2)$** 1 mark

Sum of cubes: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ with $a = 3m$, $b = 4n$, giving $(3m + 4n)(9m^2 - 12mn + 16n^2)$. Rule of signs: the bracket sign is the same as in the question, the middle sign of the trinomial is the opposite.

Source in her textbook: Unit 4, factorization of $a^3 + b^3$ — SLO: factorize expressions of the type $a^3 \pm b^3$. *Re-test of exam-15 Q13 and exam-16 Q11 (sign pattern).*

Q5. The factors of $27 - 8y^3$ are:

Correct answer: **A) $(3 - 2y)(9 + 6y + 4y^2)$** 1 mark

Difference of cubes: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ with $a = 3$, $b = 2y$, giving $(3 - 2y)(9 + 6y + 4y^2)$. Compare with Q4: same rule of signs.

Source in her textbook: Unit 4, factorization of $a^3 - b^3$ — SLO: factorize expressions of the type $a^3 \pm b^3$.

Q6. $|-9| - |-4|$ is equal to:

Correct answer: **B) 5** 1 mark

Absolute value is distance from zero, so it is never negative: $|-9| = 9$ and $|-4| = 4$. Then $9 - 4 = 5$. The bars are not brackets.

Source in her textbook: Unit 5, absolute value (" $|5| = 5$ and $|-5| = 5$ ") — SLO: define absolute value. *Re-test of exam-17 Q7 ($|-8| + |3|$ answered as -5).*

Q7. The solution of $-4 \leq 2x - 2 \leq 6$ is:

Correct answer: **C) $-1 \leq x \leq 4$** 1 mark

Add 2 to all three parts: $-2 \leq 2x \leq 8$. Divide all three parts by 2: $-1 \leq x \leq 4$. Keep the minus sign on the lower bound.

Source in her textbook: Unit 5, solving linear inequalities with rational coefficients — SLO: solve linear inequalities. *Re-test of exam-17 Q8 (minus sign dropped on the lower bound).*

Q8. If $x - 3 < 5$, then $x < 8$. The property of inequality used is the:

Correct answer: **A) additive property** 1 mark

The same number (3) is **added** to both sides without changing the direction of the inequality, so this is the **additive property**. Transitive is the chain rule ($a < b$ and $b < c \Rightarrow a < c$).

Source in her textbook: Unit 5, properties of inequalities — SLO: recognize properties of inequalities (trichotomy, transitive, additive and multiplicative). *Re-test of exam-16 Q18 (answered transitive).*

Q9. If the ordered pairs $(2a - 1, b + 3)$ and $(9, 5)$ are equal, then the value of a is:

Correct answer: **D) 5** 1 mark

Equal ordered pairs have equal corresponding elements: $2a - 1 = 9 \Rightarrow 2a = 10 \Rightarrow a = 5$ (and $b + 3 = 5 \Rightarrow b = 2$). 9 is the value of the whole element, not of a .

Source in her textbook: Unit 7, Miscellaneous Exercise 7 (equality of ordered pairs) — SLO: use conventions for coordinates in the Cartesian plane. *Re-test of exam-16 Q22 (answered with the element instead of a).*

Q10. The mid-point of the line segment joining $(-5, 4)$ and $(9, -8)$ is:

Correct answer: **B) (2, -2)** 1 mark

Mid-point = $((x_1 + x_2)/2, (y_1 + y_2)/2) = ((-5 + 9)/2, (4 + (-8))/2) = (4/2, -4/2) = (2, -2)$. $(7, -6)$ is half the *difference*, and $(4, -4)$ forgets to divide by 2.

Source in her textbook: Unit 7, mid-point formula — SLO: derive mid-point formula and calculate mid-point of a line segment. *Re-test of exam-17 Q10.*

Q	1	2	3	4	5	6	7	8	9	10
Key	B	A	C	D	A	B	C	A	D	B

SECTION B — Answers to Short Questions ($6 \times 4 = 24$ Marks)

Q.2(i)(a) Simplify $\sqrt{63} + \sqrt{28} - \sqrt{7}$ 2 marks

$$\sqrt{63} = \sqrt{9 \times 7} = 3\sqrt{7}, \quad \sqrt{28} = \sqrt{4 \times 7} = 2\sqrt{7} \quad 1$$

$$3\sqrt{7} + 2\sqrt{7} - \sqrt{7} = 4\sqrt{7} \quad 1$$

Q.2(i)(b) Evaluate $(27)^{2/3} \times (9)^{-1/2}$ 2 marks

$$(27)^{2/3} = (3^3)^{2/3} = 3^2 = 9, \quad (9)^{-1/2} = (3^2)^{-1/2} = 3^{-1} = 1/3 \quad 1$$

$$9 \times 1/3 = 3 \quad 1$$

Source in her textbook: Unit 1, radicals and laws of exponents — SLOs: concept of radicals; transform radical and exponential forms; apply the laws of exponents.

Q.2(ii)(a) Evaluate $\log 72$, naming the law at each step 3 marks

$$72 = 8 \times 9 = 2^3 \times 3^2 \quad 1$$

$$\log 72 = \log 2^3 + \log 3^2 \quad (\text{product law}) = 3 \log 2 + 2 \log 3 \quad (\text{power law}) \quad 1$$

$$= 3(0.3010) + 2(0.4771) = 0.9030 + 0.9542 = 1.8572 \quad 1$$

Marking note: the second mark needs both laws **named**. Correct working with no law named scores 2/3. Accept 72 written as $2 \times 2 \times 2 \times 3 \times 3$ with the product law named.

Q.2(ii)(b) Characteristic of $\log 0.0567$ 1 mark

$0.0567 = 5.67 \times 10^{-2}$, so the characteristic is **-2**, written **2̄** ("bar 2"). Accept either form, as in her book's Example 9 ("characteristic = -2 or 2̄"). 1

Source in her textbook: Unit 2, laws of logarithm; characteristic and mantissa (Example 9) — SLOs: prove and apply the four basic laws; define characteristic and mantissa.

Q.2(iii)(a) Factorize $1 - 343z^3$ 2 marks

$$1 - 343z^3 = (1)^3 - (7z)^3 \quad 1$$

$$= (1 - 7z)(1 + 7z + 49z^2) \quad 1$$

Q.2(iii)(b) Factorize $x^3 - 6x^2y + 12xy^2 - 8y^3$ 2 marks

$$= (x)^3 - 3(x)^2(2y) + 3(x)(2y)^2 - (2y)^3, \text{ which is } a^3 - 3a^2b + 3ab^2 - b^3 \text{ with } a = x, b = 2y \quad 1$$

$$= (x - 2y)^3 \quad 1$$

Source in her textbook: Unit 4, factorization of $a^3 - b^3$ and $a^3 - 3a^2b + 3ab^2 - b^3$ — SLO: factorize expressions of these types.

Q.2(iv)(a) Solve $|x - 3| + 4 = 9$ and verify both solutions 2 marks

$$|x - 3| = 5 \Rightarrow x - 3 = 5 \text{ or } x - 3 = -5 \Rightarrow x = 8 \text{ or } x = -2 \quad 1$$

$$\text{Check: } |8 - 3| + 4 = 5 + 4 = 9 \checkmark \quad |-2 - 3| + 4 = |-5| + 4 = 9 \checkmark \quad 1$$

Marking note: the second mark is for the verification. Both roots with no check scores 1/2.

Q.2(iv)(b) Solve $3(x - 1) \geq 2x + 4$, stating the property used 2 marks

$$3x - 3 \geq 2x + 4 \text{ (distributive law)}$$

$$3x - 3 - 2x \geq 2x + 4 - 2x \Rightarrow x - 3 \geq 4 \text{ (additive property)}$$

$$x - 3 + 3 \geq 4 + 3 \Rightarrow x \geq 7 \text{ (additive property)} \quad 1$$

Property named correctly 1. Correct answer with no property named scores 1/2.

Source in her textbook: Unit 5, absolute value equations; properties of inequalities — SLOs: solve the equation involving absolute value; recognize properties of inequalities; solve linear inequalities.

Q.2(v) Show that A(-2, -1), B(1, 3), C(7, 11) are collinear 4 marks

$$AB = \sqrt{[(1 + 2)^2 + (3 + 1)^2]} = \sqrt{(9 + 16)} = 5 \quad 1$$

$$BC = \sqrt{[(7 - 1)^2 + (11 - 3)^2]} = \sqrt{(36 + 64)} = 10 \quad 1$$

$$AC = \sqrt{[(7 + 2)^2 + (11 + 1)^2]} = \sqrt{(81 + 144)} = \sqrt{225} = 15 \quad 1$$

$$AB + BC = 5 + 10 = 15 = AC. \text{ Hence A, B and C are collinear.} \quad 1$$

Marking note: the fourth mark is for the written conclusion.

Source in her textbook: Unit 7, distance formula and collinear points (Example 3) — SLOs: find distance between two points; represent and identify collinear points.

Q.2(vi) Simplify $1/(x - 2) - 4/(x^2 - 4)$ 4 marks

$$x^2 - 4 = (x - 2)(x + 2), \text{ so the LCM of the denominators is } (x - 2)(x + 2) \quad 1$$

$$= [(x + 2) - 4] / [(x - 2)(x + 2)] \quad 1$$

$$= (x - 2) / [(x - 2)(x + 2)] \quad 1$$

$$= 1/(x + 2) \quad 1$$

Source in her textbook: Unit 4, use of HCF and LCM to reduce fractional expressions involving + and - — SLO: use HCF and LCM to reduce fractional expressions.

SECTION C — Answers to Long Questions ($2 \times 8 = 16$ Marks)

Q.3(i)(a) Square root of $x^4 - 8x^3 + 22x^2 - 24x + 9$ by division 4 marks

$$\begin{array}{r}
 x^2 - 4x + 3 \\
 \hline
 x^4 \\
 \hline
 2x^2 - 4x \\
 \hline
 2x^2 - 8x + 3 \\
 \hline
 6x^2 - 24x + 9 \\
 \hline
 6x^2 - 24x + 9 \\
 \hline
 0
 \end{array}$$

First term x^2 1, second term $-4x$ from $-8x^3 \div 2x^2$ 1, third term $+3$ with zero remainder 1

$$\therefore \text{Square root} = \pm(x^2 - 4x + 3) \quad 1$$

Marking note: the last mark needs the $\pm$, as in her book's Unit 4 Examples 29 and 30. Check: $(x^2 - 4x + 3)^2 = x^4 - 8x^3 + 22x^2 - 24x + 9$.

Source in her textbook: Unit 4, square root of algebraic expressions by division — SLO: find square root of algebraic expression by factorization and division.

Q.3(i)(b) HCF and LCM of $x^2 - 11x + 28$ and $x^2 - 3x - 28$, with verification 4 marks

$$x^2 - 11x + 28 = (x - 4)(x - 7) \quad x^2 - 3x - 28 = (x - 7)(x + 4) \quad 1$$

$$\text{HCF} = (x - 7) \quad 1$$

$$\text{LCM} = (x - 4)(x - 7)(x + 4) \quad 1$$

$\text{HCF} \times \text{LCM} = (x - 7) \times (x - 4)(x - 7)(x + 4) = [(x - 4)(x - 7)] \times [(x - 7)(x + 4)] = (x^2 - 11x + 28)(x^2 - 3x - 28)$. Hence **HCF $\times$ LCM = product.** 1

Source in her textbook: Unit 4, HCF and LCM by factorization; relationship between HCF and LCM — SLOs: find HCF and LCM; know the relationship between HCF and LCM.

Q.3(ii) Lamp post between houses P(-4, 3) and Q(6, -7) 8 marks

(a) $M = ((-4 + 6)/2, (3 + (-7))/2) = (2/2, -4/2) = M(1, -2)$ 2

(b) $PM = \sqrt{[(1 + 4)^2 + (-2 - 3)^2]} = \sqrt{(25 + 25)} = \sqrt{50} = 5\sqrt{2}$ 1

$QM = \sqrt{[(1 - 6)^2 + (-2 + 7)^2]} = \sqrt{(25 + 25)} = 5\sqrt{2}$. Hence $PM = QM$. 1

(c) $RM = \sqrt{[(8 - 1)^2 + (-1 + 2)^2]} = \sqrt{(49 + 1)} = \sqrt{50} = 5\sqrt{2}$ 1

$RM = PM = QM = 5\sqrt{2}$, so **R is at the same distance from the lamp post as P and Q.** 1

(d) $PQ = \sqrt{[(6 + 4)^2 + (-7 - 3)^2]} = \sqrt{(100 + 100)} = \sqrt{200} = 10\sqrt{2}$ 1

$PM + MQ = 5\sqrt{2} + 5\sqrt{2} = 10\sqrt{2} = PQ$. Hence **P, M and Q are collinear** (M lies on the line segment PQ). 1

Marking note: marks in (b), (c) and (d) include the written conclusion. In (b), $\sqrt{50}$ left unsimplified loses 0.5, since the question asks for simplest radical form.

Source in her textbook: Unit 7, distance and mid-point formulas; the unit opener (a lamp post equidistant from two homes) — SLOs: find distance between two points; calculate mid-point; represent collinear points; apply distance and mid-point formulas to solve real life situations.

— END OF ANSWER KEY —